

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

TITLE OF THE PROJECT	Handwritten Digit/Character Recognition	
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ABSTRACT

Recognizing handwritten digits and characters is one of those problems that sounds simple but turns out to be surprisingly tricky once you get into it — no two people write the same digit exactly the same way, and even the same person's handwriting changes from one attempt to the next. In this project, we look at two classical approaches, **K-Nearest Neighbors (KNN)** and **Support Vector Machine (SVM)**, and compare them against a basic **neural network** to see which one does a better job at correctly identifying handwritten digits and characters.

We used the **MNIST** dataset for digits (with **EMNIST** considered for extending the work to alphabetic characters), which contains thousands of labeled grayscale images of handwritten samples. Before feeding this data into any model, the images were normalized and resized, and in some cases we extracted **HOG (Histogram of Oriented Gradients)** features to help the classical models perform better, since **KNN** and **SVM** don't automatically learn features the way a neural network does. The **KNN** model works by comparing a new digit to its closest neighbors in the training data and picking the most common label among them. Alongside these, a simple **neural network** with one or two hidden layers was trained using **backpropagation** to see how well it could learn patterns directly from the raw pixel values without needing hand-crafted features.

Once all three models were trained, they were evaluated on the same test data using **accuracy**, **precision**, **recall**, **F1-score**, and **confusion matrices**, along with a rough comparison of how long each model took to train and predict. As expected, the **neural network** came out ahead in terms of overall accuracy, since it can pick up on subtle patterns in the pixel data that the other two models simply can't capture. That said, **KNN** and **SVM** weren't far behind and have their own advantages — they're easier to understand, quicker to set up, and don't need as much computing power, which matters if you're working with limited resources or want a model that's easy to explain.

Overall, this project shows that while the **neural network** is the stronger performer here, the classical methods still hold up reasonably well and are worth considering depending on the situation. It also sets up a natural next step for future work — exploring **Convolutional Neural Networks (CNNs)**, which are built specifically for image data and would likely push accuracy even further.

Signature of the Guide

HOD